Day 140 | Unit 10 | Answer key

AASA Applications and Cumulative Review

Course Final Test

Standards: 6.NS.B.3, 6.EE.A.1, 6.RP.A.1, 6.NS.C.5, 6.EE.A.2b, 6.SP.A.1, 6.NS.B.2, 6.EE.B.7, 6.RP.A.3a, 6.EE.A.2c, 6.G.A.1, 6.EE.A.2a, 6.EE.B.6, 6.NS.A.1, 6.RP.A.2, 6.RP.A.3b, 6.EE.A.3, 6.EE.A.4, 6.NS.B.4a, 6.SP.B.4, 6.SP.B.5c, 6.NS.C.7a, 6.NS.C.7b, 6.RP.A.3d, 6.NS.C.7c, 6.NS.C.7d, 6.RP.A.3c, 6.G.A.4, 6.EE.B.5, 6.NS.C.6b, 6.NS.C.6c, 6.EE.B.8, 6.G.A.2, 6.NS.C.8, 6.EE.C.9, 6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5d

Learning target: I can apply and explain Grade 6 mathematics across the full course.

1. 47.7

Align decimal points; 39.5 + 8.2 = 47.7.

6.NS.B.3 | Constructed response

2. 4 × 4 × 4 = 64

The exponent counts 3 factors of 4; multiply all factors.

6.EE.A.1 | Constructed response

3. 1:1

Keep red first; divide 5:5 by 5.

6.RP.A.1 | Constructed response

4. -20 m; 0 means at sea level

Sea level is the reference. Below is negative and above is positive.

6.NS.C.5 | Constructed response

5. Terms 2x, 7y, 4; coefficients 2, 7; constant 4

Terms are separated by addition; coefficients multiply variables.

6.EE.A.2b | Constructed response

6. B, D - How does mass vary among the dogs?; What is the mass in kilograms of each of the dogs?

Different dogs can have different mass. The questions anticipate variation across a collection. The conversion and arithmetic questions have fixed answers.

6.SP.A.1 | Multiple select

7. 60

30 ÷ 5: write 6, subtract 30, remainder 0; bring down the next digit: 0 ÷ 5: write 0, subtract 0, remainder 0. Check: 5 × 60 = 300.

6.NS.B.2 | Constructed response

8. x = 4

Subtract 2 on both sides. Then x = 4; substitution gives 6 on each side.

6.EE.B.7 | Constructed response

9. B - 97/100

Align decimal points; 2.6 - 1.63 = 0.97.

6.NS.B.3 | Multiple choice

10. C, D - (6, 4); (9, 6)

Equivalent ratios multiply both quantities by the same factor. A uses 2; B uses 3. C and D change only one scaled quantity.

6.RP.A.3a | Multiple select

11. 17

Parentheses: 5 - 3 = 2. Exponent: $3^2 = 9$. Multiply: 4 × 2 = 8. Add: 9 + 8 = 17. The product must be completed before the final sum.

6.EE.A.1, 6.EE.A.2c | Numeric entry and error analysis

12. C - 336/125

Compute the whole-number product and restore decimal places: 1.92 × 1.4 = 2.688.

6.NS.B.3 | Multiple choice

13. B, C - 7 × 5; (14 × 5) ÷ 2

Triangle area is half the matching parallelogram: (14 × 5)/2 = 35 cm^2. Multiplying half the base by the height gives the same result.

6.G.A.1 | Multiple select

14. 5x + 2y - 2

Marker cost is 5x; folder cost is 2y. Add these subtotals, then subtract the one fixed coupon of 2. An expression does not need an equals sign.

6.EE.A.2a, 6.EE.B.6 | Expression entry and reasoning

15. D - 11/2

Multiply both numbers by 10: 137.5 ÷ 25 = 5.5. Check: 2.5 × 5.5 = 13.75.

6.NS.B.3 | Multiple choice

16. Aligned pairs (0,0), (3,3), (6,6), (21,21); 21 m^2

Each aligned jump adds 3 liters and 3 m^2. Both quantities scale by 7; the number lines measure different units.

6.RP.A.3a | Double number line and explanation

17. 39/2

Substitute: 7 × (3/2) + 3². The product is 21/2 and the square is 9. Add to get 39/2.

6.EE.A.2c | Numeric entry and reasoning

18. B - 2

Each small part is 1/4; 2 shaded parts divided into groups of 1 gives 2/1 = 2 groups.

6.NS.A.1 | Multiple choice

19. C, D - 3 notebooks cost $15.; Each notebook costs $5.

Cost per notebook = 25 ÷ 5 = 5. Multiply $5 by the chosen notebook count.

6.RP.A.2, 6.RP.A.3b | Multiple select

20. B, C - 2x + 8; 8 + 2x

Distribute 2 to both terms to get 2x + 8. Reversing the addends gives the same value. The other expressions fail, for example at x = 0.

6.EE.A.3 | Multiple select

21. D - 35/12

7/4 ÷ 3/5 = 7/4 × 5/3 = 35/12.

6.NS.A.1 | Multiple choice

22. Removed 12 m²; remaining 88 m²

Outer area = 100. Triangle = 4 × 6/2 = 12. Subtract the removed region exactly once.

6.G.A.1 | Two-part numeric entry

23. C, D - (5 + 6)x + 5; 11x + 5

Combine coefficients of like x terms: 5 + 6 = 11. The constant 5 stays separate.

6.EE.A.3 | Multiple select

24. A - 2

5 ÷ 5/2 = 5 × 2/5 = 2.

6.NS.A.1 | Multiple choice

25. A: $4 each; B: $5 each; A is the better buy.

Divide each cost by its own notebook count: 24/6 = 4; 35/7 = 5. Compare the same one-notebook unit.

6.RP.A.3b | Two-part numeric entry and reasoning

26. B, D - 6x + 24; 24 + 6x

Distribute 6 to both terms to get 6x + 24. Reversing the addends gives the same value. The other expressions fail, for example at x = 0.

6.EE.A.4 | Multiple select

27. A, D - 6 is the largest number of identical no-leftover kits from 12 pencils and 18 erasers.; 6 divides both 12 and 18.

GCF(12, 18) = 6. Divide each count by 6: 2 pencils and 3 erasers per kit. A larger shared divisor is impossible.

6.NS.B.4a | Multiple select

28. (5, 8, 19/2, 11, 14); IQR 3 min

Minimum and maximum anchor whiskers. Box ends are 8, 11; median line is 19/2. IQR = 11 - 8.

6.SP.B.4, 6.SP.B.5c | Box plot and numeric entry

29. A, B - 18/4; 9/2

Use the inverse operation on both sides. The solution is 9/2; the doubled numerator and denominator name the same value. Substitution verifies the solution.

6.EE.B.7 | Multiple select

30. B, D - In 2 hours it travels 40 km.; The speed is 20 km per hour.

Divide distance by time: 40 ÷ 2 = 20 km/h. Multiply this unit rate by 2 for the later distance.

6.RP.A.3b | Multiple select

31. B, C - -9 < -7; -7 lies to the right of -9 on a number line.

Among negatives, the number closer to zero is greater. Thus -7 is warmer and is farther right.

6.NS.C.7a, 6.NS.C.7b | Multiple select

32. A, D - 14/4; 7/2

Use the inverse operation on both sides. The solution is 7/2; the doubled numerator and denominator name the same value. Substitution verifies the solution.

6.EE.B.7 | Multiple select

33. A, C - 42 inches; 7/2 × 12 inches

There are 12 smaller units in each larger one. Multiply 7/2 by 12, giving 42 inches. The physical length does not change.

6.RP.A.3d | Multiple select

34. A, D - -9 is farther from zero than 7.; |-9| = 9

Absolute value measures distance: 9 versus 7. A larger distance does not make the negative number greater.

6.NS.C.7c, 6.NS.C.7d | Multiple select

35. A, C - 26/4; 13/2

Use the inverse operation on both sides. The solution is 13/2; the doubled numerator and denominator name the same value. Substitution verifies the solution.

6.EE.B.7 | Multiple select

36. 12 participants; n = (15/100) × 80; less than the whole because the percent is below 100

Rename 15% as 3/20. Multiply by 80 to obtain 12. Compare the rate with 1.

6.RP.A.3c | Numeric entry and equation

37. 72 cm²; open box 60 cm²

Closed area = 2(12+18+6) = 72. Subtract exactly one top face, 12, for the open box.

6.G.A.4 | Two-part numeric entry

38. 16 observations; highest intervals 10–14

Use four equal-width touching bars with heights 3, 4, 5, 4. Frequencies sum to 16. Include every interval with frequency 5.

6.SP.B.4 | Data display and explanation

39. 1 bottles; 2/3 L left

2 ÷ 4/3 = 2 × 3/4 = 3/2. Only whole bottles count; subtract the liquid used.

6.NS.A.1 | Model and constructed response

40. A, D - 5; 4

Subtract 2 conceptually: x must exceed 3. Equality at 3 fails because the inequality is strict.

6.EE.B.5 | Multiple select

41. A, C - (125/100)w = 75; w = 75 ÷ (125/100)

The part is percent rate times whole. Divide 75 by 5/4; w = 60. Check by multiplication.

6.RP.A.3c | Multiple select

42. P: II; Q: IV; R is on the y-axis.

P has signs (-,+), so II. Q has (+,-), so IV. R has x = 0 and is on an axis.

6.NS.C.6b, 6.NS.C.6c | Coordinate plotting and explanation

43. A, C - n ≥ 4; 4 is an allowed membership count.

At least includes the boundary, so n ≥ 4. On a real number line use a closed point and shade right; actual member counts are whole numbers.

6.EE.B.8 | Multiple select

44. 65/4 m³; 130 cubes

Multiply edges: 13/2 × 5/2 × 1 = 65/4. The edge counts are 13, 5, 2, giving 130 cubes. Each cube is 1/8 m³; 130/8 = 65/4.

6.G.A.2 | Numeric entry and model reasoning

45. 20%; denominator 100

Percent = (20 ÷ 100) × 100 = 20%. The denominator is the reference whole, not automatically the larger number.

6.RP.A.3c | Numeric entry and explanation

46. AB = 5 units; AC = 6 units

AB is horizontal: 2 + 3 = 5. AC is vertical: 3 + 3 = 6. Each segment crosses an axis.

6.NS.C.8 | Coordinate plotting and numeric entry

47. (0,0), (1,4), (2,8); independent h; L = 4h

Hours are the input. Multiply each hour value by 4 to obtain liters. Graph hours horizontally and liters vertically.

6.EE.C.9 | Table completion, equation, and graph

48. 6 observations; minutes; median 11, IQR 6; mean 37/3, MAD 19/3. A high value makes median/IQR a useful choice.

Order the observations. Middle positions give median 11 and half medians give IQR 6. Mean is 74/6; averaging distances gives MAD 19/3. Tie the chosen summary to the displayed shape and context.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d | Statistical reasoning and explanation